Computer Science (2024-25)

CLASS XII Code No. 083 (Practical Practice Paper)

Time allowed: 3 Hrs

S.No	Unit Name	Marks (Total=30)
1	Lab Test: 1. Python program (60% logic + 20% documentation + 20% code quality)	8
	2. SQL queries (4 queries based on one or two tables)	4
2	Report file: • Minimum 15 Python programs. • SQL Queries – Minimum 5 sets using one table / two tables. • Minimum 4 programs based on Python – SQL connectivity	7
3	Project (using concepts learnt in Classes 11 and 12)	8
4	Viva voce	3

1.	**Write a Menu Driven Python Program to perform PUSH, POP and PEEP operation on stack (Price) implemented as List.** **1. PUSH** **2. POP** **3. DISPLAY** **4. EXIT** **CODE:** stack = [] def push(): price = float(input("Enter the price to push: ")) stack.append(price) print("Price pushed successfully!")	8

	<pre>def pop(): if not stack: print("Stack is empty!") else: print(f"Popped price: {stack.pop()}") def display(): if not stack: print("Stack is empty!") else: print("Stack contents:", stack) while True: print("\n1. PUSH\n2. POP\n3. DISPLAY\n4. EXIT") choice = int(input("Enter your choice: ")) if choice == 1: push() elif choice == 2: pop() elif choice == 3: display() elif choice == 4: print("Exiting...") break else: print("Invalid choice!")</pre>	
2.	**Complete the given Python program to search for an employee number in a table emp.** **Following parameters are required for the connection:** **Server name: localhost** **User: root** **Password: admin** **Database: company** **import mysql.connector as mycon** **cnt = mycon.connect(____________________________________) # Statement 1** **mycursor = cnt._____________ # Statement 2**	4

	empno = int(input('Enter Employee Number :')) **query = ____ # Statement 3** **mycursor.execute(query)** **record = mycursor.____________ # Statement 4** **for r in record:** **print(r)** **CODE:** import mysql.connector as mycon cnt = mycon.connect(host="localhost", user="root", password="admin", database="company") # Statement 1 mycursor = cnt.cursor() # Statement 2 empno = int(input('Enter Employee Number: ')) query = f"SELECT * FROM emp WHERE empno = {empno}" # Statement 3 mycursor.execute(query) record = mycursor.fetchall() # Statement 4 for r in record: print(r)	
3.	**Write a menu-driven Python program to perform the following operations on a binary file student.dat:** **1. Add New Student** **2. Show All Students** **3. Search Student** **4. Exit** **(The binary file student.dat contains [roll, name, per])** **CODE:** import pickle filename = "student.dat" def add_student(): with open(filename, "ab") as file: roll = int(input("Enter Roll Number: ")) name = input("Enter Name: ")	

```
        per = float(input("Enter Percentage: "))
        pickle.dump([roll, name, per], file)
        print("Student added successfully!")

def show_students():
    try:
        with open(filename, "rb") as file:
            while True:
                student = pickle.load(file)
                print(student)
    except EOFError:
        pass

def search_student():
    roll = int(input("Enter Roll Number to Search: "))
    found = False
    try:
        with open(filename, "rb") as file:
            while True:
                student = pickle.load(file)
                if student[0] == roll:
                    print("Student Found:", student)
                    found = True
                    break
    except EOFError:
        if not found:
            print("Student not found.")

while True:
    print("\n1. Add New Student\n2. Show All Students\n3. Search Student\n4.
Exit")
    choice = int(input("Enter your choice: "))
    if choice == 1:
        add_student()
    elif choice == 2:
        show_students()
    elif choice == 3:
        search_student()
    elif choice == 4:
        print("Exiting...")
```

	`` break else: print("Invalid choice!") ```	
4.	**Complete the given Python program to add new employee details in a table emp.** **Following parameters are required for the connection:** **Server name: localhost** **User: root** **Password: admin** **Database: company** **import mysql.connector as mycon** **cnt = mycon.connect(________________________________) # Statement 1** **mycursor = cnt._____________ # Statement 2** **empno = int(input('Enter Employee Number :'))** **name = input('Enter Name :')** **dept = input('Enter Department:')** **salary = int(input('Enter Salary :'))** **mycursor.execute(________________________________) # Statement 3** **cnt._____________ # Statement 4** **CODE:** import mysql.connector as mycon cnt = mycon.connect(host="localhost", user="root", password="admin", database="company") # Statement 1 mycursor = cnt.cursor() # Statement 2 empno = int(input("Enter Employee Number: ")) name = input("Enter Name: ") dept = input("Enter Department: ") salary = int(input("Enter Salary: ")) query = f"INSERT INTO emp (empno, name, dept, salary) VALUES ({empno}, '{name}', '{dept}', {salary})" # Statement 3 mycursor.execute(query)	

	cnt.commit() # Statement 4 print("Employee added successfully!")	
5.	**Write a menu-driven Python program to perform the following operations on a CSV file book.csv:** **1. Add New Book** **2. Show All Books** **3. Delete Book** **4. Exit** **(The CSV file book.csv contains [bookno, bname, author, price])**	

CODE:

```
import csv

filename = "book.csv"

def add_book():
    with open(filename, "a", newline='') as file:
        writer = csv.writer(file)
        bookno = int(input("Enter Book Number: "))
        bname = input("Enter Book Name: ")
        author = input("Enter Author Name: ")
        price = float(input("Enter Price: "))
        writer.writerow([bookno, bname, author, price])
        print("Book added successfully!")

def show_books():
    try:
        with open(filename, "r") as file:
            reader = csv.reader(file)
            for row in reader:
                print(row)
    except FileNotFoundError:
        print("No books found.")

def delete_book():
    bookno = int(input("Enter Book Number to Delete: "))
    rows = []
    found = False
    with open(filename, "r") as file:
```

```
        reader = csv.reader(file)
        for row in reader:
            if int(row[0]) != bookno:
                rows.append(row)
            else:
                found = True
    with open(filename, "w", newline='') as file:
        writer = csv.writer(file)
        writer.writerows(rows)
    if found:
        print("Book deleted successfully!")
    else:
        print("Book not found.")

while True:
    print("\n1. Add New Book\n2. Show All Books\n3. Delete Book\n4. Exit")
    choice = int(input("Enter your choice: "))
    if choice == 1:
        add_book()
    elif choice == 2:
        show_books()
    elif choice == 3:
        delete_book()
    elif choice == 4:
        print("Exiting...")
        break
    else:
        print("Invalid choice!")
```

6. **Complete the given Python program to delete a book number in a table books.**

Following parameters are required for the connection:

Server name: localhost

User: root

Password: admin

Database: library

```
import mysql.connector as mycon

cnt = mycon.connect(________________________________)  # Statement 1
```

	mycursor = cnt.______________ # Statement 2 **bookno = int(input('Enter Book Number :'))** **mycursor.execute(__________________________________) # Statement 3** **cnt.______________ # Statement 4** **CODE:** import mysql.connector as mycon cnt = mycon.connect(host="localhost", user="root", password="admin", database="library") # Statement 1 mycursor = cnt.cursor() # Statement 2 bookno = int(input("Enter Book Number: ")) query = f"DELETE FROM books WHERE bookno = {bookno}" # Statement 3 mycursor.execute(query) cnt.commit() # Statement 4 print("Book deleted successfully!")	
7.	**Write a menu-driven Python program to manage student records in a text file named students.txt. The program should perform the following operations:** 1. **Add a New Student: Append a new student's details (roll number, name, percentage) to the file.** 2. **Show All Students: Read and display all student records stored in the file.** 3. **Search for a Student: Search for a student by their roll number and display their details if found.** 4. **Exit: Terminate the program.** **Note: Each record in the file should be stored in the format:** **roll number, name, percentage** **CODE:** filename = "students.txt" def add_student(): """Add a new student to the text file.""" roll = input("Enter Roll Number: ") name = input("Enter Name: ") percentage = input("Enter Percentage: ")	

```
    with open(filename, "a") as file:
        file.write(f"{roll}, {name}, {percentage}\n")
    print("Student added successfully!")

def show_students():
    """Display all students' details from the text file."""
    try:
        with open(filename, "r") as file:
            print("\n--- Student Details ---")
            for line in file:
                roll, name, percentage = line.strip().split(", ")
                print(f"Roll: {roll}, Name: {name}, Percentage: {percentage}")
    except FileNotFoundError:
        print("No student records found. Add some students first.")

def search_student():
    """Search for a student by roll number."""
    roll_to_search = input("Enter Roll Number to Search: ")
    found = False
    try:
        with open(filename, "r") as file:
            for line in file:
                roll, name, percentage = line.strip().split(", ")
                if roll == roll_to_search:
                    print(f"Student Found: Roll: {roll}, Name: {name}, Percentage:
{percentage}")
                    found = True
                    break
        if not found:
            print("Student not found.")
    except FileNotFoundError:
        print("No student records found. Add some students first.")

def main():
    """Menu-driven program for text file operations."""
    while True:
        print("\n1. Add New Student\n2. Show All Students\n3. Search
Student\n4. Exit")
        choice = input("Enter your choice: ")
```

	`` if choice == "1": add_student() elif choice == "2": show_students() elif choice == "3": search_student() elif choice == "4": print("Exiting the program. Goodbye!") break else: print("Invalid choice. Please try again.") # Run the program main() ```	
8.	**Complete the given Python program to update a student's percentage in the students table.** **Following parameters are required for the connection:** **Server Name: localhost** **User: root** **Password: admin** **Database: school** **import mysql.connector as mycon** **cnt = mycon.connect(_________________________________) # Statement 1** **mycursor = cnt._____________ # Statement 2** **roll_no = int(input("Enter Roll Number: "))** **new_percentage = float(input("Enter New Percentage: "))** **query = _________________________________ # Statement 3** **mycursor.execute(query, (________________)) # Statement 4** **cnt._____________ # Statement 5** **print("Student percentage updated successfully!")** **Code:** import mysql.connector as mycon	

```
cnt = mycon.connect(host="localhost", user="root", password="admin",
database="school")  # Statement 1
mycursor = cnt.cursor()  # Statement 2

roll_no = int(input("Enter Roll Number: "))
new_percentage = float(input("Enter New Percentage: "))

query = "UPDATE students SET percentage = %s WHERE roll_no = %s"  #
Statement 3
mycursor.execute(query, (new_percentage, roll_no))  # Statement 4
cnt.commit()  # Statement 5

print("Student percentage updated successfully!")
```